RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie
....
MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR
....
COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN HIGHER NATIONAL DIPLOMA EXAM (HND)
....

REPUBLIC OF CAMEROON
Peace-Work-Fatherland
....
MINISTRY OF HIGHER EDUCATION
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NATIONAL COMMISSION FOR THE ORGANISATION OF HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM
....

National Exam of Higher National Diploma 2022 Session

Spécialty/option: Software Engineering (SWE) **Credits : 14**
Paper: Cass Study
Duration : 6 Hours

GENERAL INSTRUCTIONS.
You are reminded of the necessity of good English and orderly presentation of your material. Where calculations are required, clearly show your working and be chronological in your answer.

SECTION A: ALGORITHM AND PROGRAMMING. (50MARKS)

I. Algorithms (10Marks)

1- Explain what is Space complexity of an algorithm means. **(2marks)**

2- An algorithm execution time can be said to be worst case, average case, or best case. Define these notations **3x1=3marks)**

3- Write an algorithm that receives three numbers and displays the maximum of these numbers using

i. Pseudocode

ii. Flowcharts **(2.5x2=5marks)**

II. Procedural Programming (20Marks)

1- A class of n students take an annual examination in m subjects. Two of such subjects are coefficient 4 subjects and the rest are coefficient 2. Mathematics and English are the two coefficient 4 subjects and they are compulsory. Each student must offer between 4 and 10 subjects. Write a c program that permits the user to

i. To enter the number of subjects written, marks on twenty and the coefficient of each of the subjects written.

ii. The program will print out the total marks of the student, his average and his remark (0<=average<=6: ver ypoor, 6<average<=9: poor, 9<average<10: belowaverage, average=10: average, average>10 :good **(10marks)**

2- Write a function named "location_of_target" that takes as its arguments the following:

(1) an array of integer values;

(2) an integer that tells how many integer values are in the array;

(3) an integer "target value".

The function should determine whether the given target value occurs in any of the cells of the array, and if it does, the function should return the subscript of the cell containing the target value. If more than one of the cells contains the target value, then the function should return the largest subscript of the cells that contain the target value.

If the target value does not occur in any of the cells, then the function should return the sentinel value -1. Thus, for example, if the target value that's passed to the function is 34 and the array that's passed to the function looks like this:

0	1	2	3	4	5	6
58	26	91	34	70	34	88

then the target value occurs in cells 3 and 5 , so the function should return the integer value 5.

(10marks)

III. Object oriented programming (20Marks)

1- Define the following as seen in object oriented programming.

i. Object

ii. Classes

iii. Methods **(3x1= 3marks)**

2- What are the four major principles that make a language to be object oriented?

(4marks)

3- What are the differences between private and public class members? **(2marks)**

4- What are the differences between data hiding and encapsulation? **(2marks)**

5- Design a class called Cube that has the following data member side of data type double. Supply the following methods: a constructor with one argument, a getter and setter method for the side of the cube, the volume equals to side raised to the power three.

Implement the class Cube and write a test class Cube Test in C++ or Java to verify the functionalities of the class Cube **(9marks)**

SECTION B: DATABASE DEVELOPMENT AND ADMINISTRATION .(20MARKS)

A private university is interested in designing a database that will track its teachers, students and courses. Information of interest includes teacher's names, matricule, telephone, gender, grade, student names, student telephone, student matricule, student gender, student address, student age, course title, course code, course ID and coursecredit. Each student enrolls in one or many courses of his choice and a teacher is teaching one or many courses. When a student enrolls to a course at the end, he had a mark.

Problems to solve:

1- What is an Entity? **(1 mark)**

2- List all the possible entities. **(1.5 marks)**

3- What is an attribute? **(1 mark)**

4- On each entity listed in question 2, list its attributes. **(1.5 marks)**

5- What is an E-R Diagram? List all the components of an E-R diagram **(3 marks)**

6- Draw the equivalent E-R diagram of that private University using the crow's notation or Chen notation. **(3 marks)**

7- Convert the diagram to a relational model **(3 marks)**

8- Using SQL language answer the following questions:

   i. Create the tables of your database **(3 marks)**

   ii. Create the database of the university called "UNIV" **(1 mark)**

   iii. Insert a student your database **(1 mark)**

   iv. Retrieve the marks of all students in a particular subject **(1 mark)**

SECTION C: WEB DESIGN. (15MARKS)

1. Draw a picture of how the following HTML/CSS code will look when the browser renders it on-screen. Assume that the HTML is wrapped in a valid full page with a head and body. Indicate a non-white background by shading lightly. It is possible that some CSS rules shown will not apply to any elements. **(7mks)**

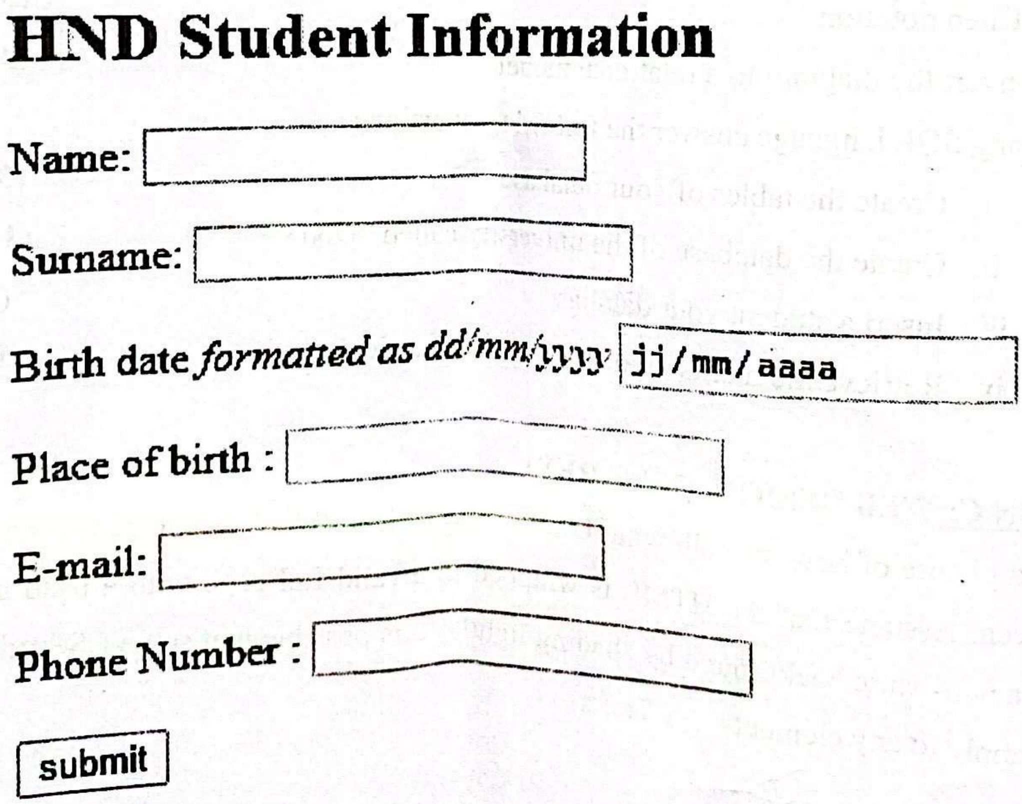
HTML:

```html
<p> hello </p>
<div>
    <div> one </div>
    <div id="div"> two <br /> two two </div>
</div>
<div class="div"> three <br /> three three <br /> three three three </div>
<div class="div"> four <br /> four four <br /> four four four <br />
                  <span class="div"> four four four four </span> </div>
<p> goodbye </p>
```

CSS:

```css
div          { background-color: white;  border: 2px dashed black;  padding: 1em; }
body > div   { float: right; }
div div      { background-color: cyan;  margin: 1em;  text-align: right; }
#div         { text-decoration: underline; }
.div .div    { border: 2px solid black; }
p            { border: 2px solid black;  clear: both; }
```

2. Give HTML5 code to create this form



(8marks)

SECTION D: NETWORKING. (15MARKS)

1- For the lower four layers of the OSI model ,give for each : (**8marks)**

   i. Functions of the layer.

   ii. An example of a protocol used in the layer

   iii. An example of network or interconnection equipment and/or end Systems used.

2- What is a network topology? (**1mark)**

3- For each of the following network give a sketch, two advantages and two disadvantages

   i. Star

   ii. Ring **(6 marks)**